

TAG is not a phrasal Construction Grammar, it is
a Transformational Grammar, or is it?
A comment on the nature of grammatical relations

Draft: Comments welcome!

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Abstract

This paper discusses recent claims about Tree Adjoining Grammar (TAG) being suited as the formalization for phrasal Construction Grammar approaches. It is shown that inheritance as such is not a sufficient means to establish relations between linguistic objects and to capture generalizations. Of course, TAG has all the technical tools that are needed for expressing the generalizations, but if they are used appropriately, TAG is more similar to a Transformational Grammar than to a Construction Grammar. This discussion helps to show where the shortcomings of phrasal Construction Grammar proposals are and what should be done instead if one wants to assume a constructionist approach.

1 Tree Adjoining Grammar as Construction Grammar?

Starting with Lakoff (1987) and Fillmore et al. (1988) the framework of Construction Grammar got more and more popular. Goldberg (1995) applied the ideas of Construction Grammar to argument structure constructions. One popular example is given in (1):

- (1) Sam sneezed the napkin off the table. (Goldberg 1995: 29, 55)

The interesting aspect about (1) is that *sneeze* is an intransitive, mono-valenced verb. Using *sneeze* with an object as in (2) is ungrammatical.

- (2) *He sneezed the napkin.

It is also interesting to note that (1) has a causative meaning. Goldberg and others argued that this meaning component is not part of the meaning of the involved lexical items and hence has to be contributed by the configuration into which the lexical items are inserted. By inserting *sneeze* into a certain configuration, we get an argument extension and additional meaning components. I call such proposals phrasal Construction Grammar in the following. Müller (2006, 2013b) and Müller & Wechsler (2014a) argued instead that all phenomena having to do with valence should be treated lexically. This implies that valence changing “processes” like passivization should be treated lexically as well. In the last decade, various papers appeared that tried to show that Tree Adjoining Grammar (Joshi et al. 1975, Joshi & Schabes 1997, Abeillé & Rambow 2000) is the perfect implementation of phrasal Construction Grammar (Lichte & Kallmeyer 2017, Kallmeyer & Osswald 2013, Seyffarth 2019, 2023).¹ This claim has some initial plausibility, since TAG has the unique property of working with tree

¹A reviewer pointed out that the title of the paper seems to be a category mistake. This criticism is warranted since TAG is a grammar formalism and it is possible to write very different grammars within this formalism. However, grammar formalisms are also often associated with sets of guiding assumptions. The word framework is often used to refer to packages of formal tools and basic assumptions. Goldberg (2003a) formulated ten tenets for Construction Grammar and many researchers working in more formal non-transformational theories try to convince Construction Grammarians that their framework is in fact constructional and hence might be adopted for doing Construction Grammar work. Examples are Constructional HPSG (Sag 1997, Müller et al. 2024, Müller 2024), Sign-Based Construction Grammar (Sag 2012), which is an HPSG variant (Sag 2010: 486), Lexical Functional Grammar (Asudeh et al. 2013, Findlay 2023). Findlay (2023) used the title *Lexical Functional Grammar as a Construction Grammar*. The title of Lichte & Kallmeyer (2017) is *Tree-Adjoining Grammar: A tree-based constructionist grammar framework for natural language understanding* and it is exactly what is described here: an explanation of why TAG is suited as a formalization of Construction Grammar. The paper refers to the tenets of Goldberg.

The point of the current paper is not to show that any of the named frameworks is not suited as a Construction Grammar. All of these frameworks can capture phrasal constructions like Jackendoff’s (2008) *student after student* construction. Rather I want

configurations that include all arguments of a head, even in the analysis of sentences involving long distance dependencies (Joshi & Schabes 1997: 99–100, Lichte & Kallmeyer 2017: 208). TAG is said to have an extended domain of locality, since elements that may be realized far away are parts of so-called elementary trees (Joshi & Schabes 1997: 95–96). Furthermore TAG has a convincing analysis of idioms like *take sth into account* with fixed material in discontinuous parts of idioms (Abeillé & Schabes 1989).

TAG has two basic modes of combination: substitution and adjunction. For example, there can be a tree for an intransitive verb with an open slot into which the subject can be inserted. For example, Figure 1 shows how the elementary tree for the NP *Kim* is inserted into the elementary tree for *laughs*.

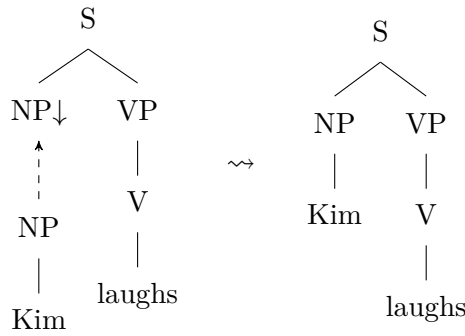


Figure 1: Substitution

Figure 2 shows an example of how the adjunction tree for *always* can be used. The adjunct provides a node that can be identified with a node in an

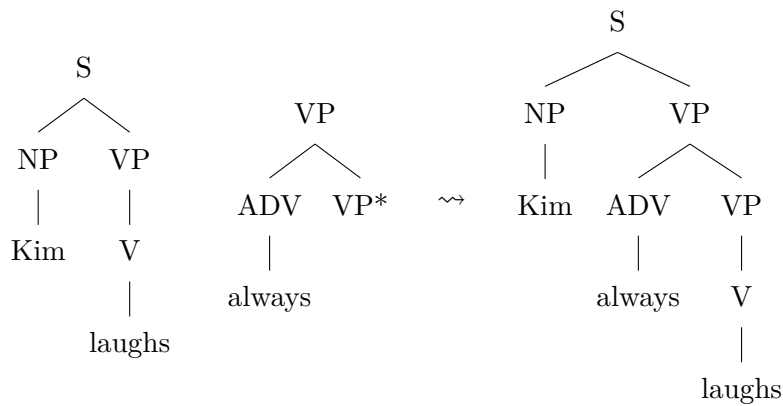


Figure 2: Adjunction

to show once more that none of them can handle argument structure construction the way it was suggested by Goldberg (1995).

already existing tree. The tree is taken apart and the adjunct is inserted into the middle. The top node of the adjunct tree and the node marked with a star fill the position where the main tree is taken apart.

The analysis of *take something into account* is provided in Figure 3. The fixed material is specified in the respective tree positions and slots for the subject and an object are still open. NA stands for no adjunction and rules out adjuncts in the middle of fixed parts.

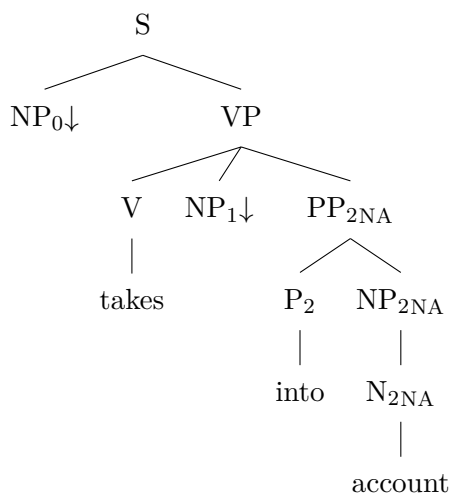


Figure 3: Analysis of a discontinuous idiom in TAG: NP_0 takes NP_1 into account. according to Abeillé & Schabes (1989: 7)

Lexical items can be used with several trees, so-called tree families. For example, there can be active and passive trees. The question is how these trees are derived or described. Starting with Candito’s (1998) work, a meta-grammar formalism is used to describe trees by using inheritance mechanisms and to capture generalizations. Since some of the tree computation is done automatically, this makes also certain predictions. However, as Müller pointed out in several publications, there are mathematical limits to what can be done with inheritance (Müller 2010, 2013a: Section 7.5.2, 2017b).

TAG has one more level for the description of language now: one level with inheritance on the meta level and the combinatoric level that has been there in TAG right from the beginning. The question is whether there is a clever combination of tools so that the shortcomings of phrasal approaches that Müller and Wechsler described in various publications can be overcome (Müller 2006, 2010, 2013b, 2018, Müller & Wechsler 2014a,b). The problem of inheritance is that it cannot model recursive embeddings. They are needed for semantics. Embedding is also needed for changes of part of speech. In phonology (Kiparsky 1968: 196–198) and Transformational Grammar (G. Müller 2017a) there were feeding cycles, input to and output from phono-

logical processes/transformations. This means that there is a rule order, something that some authors explicitly reject (Crabbé et al. 2013: 596–597). Furthermore, TAG has a problem that is similar to a problem classical Government & Binding (Chomsky 1981) has: There are fixed structures that are coupled with semantics. TAG has form-meaning pairs involving syntactic structure and GB assigns theta roles to certain tree configurations. In comparison, Categorical Grammar and HPSG have valence information, which is potential structure rather than actual structure (Müller 2019). I will show in the following that the assumption of valence information is empirically necessary and the assumption of full blown structure just gets into the way when approaching morphology.

2 The history of the debate and TAGs extended domain of locality

The discussion of whether phrasal vs. lexical constructional approaches are more appropriate started in 2006 with a reply of Müller to a paper by Goldberg & Jackendoff (2004) in *Language*. Müller pointed out that the attractive idea that there are form-meaning pairs that have to be learned is not applicable to languages like German, which have freer constituent order and partial fronting. Apart from this the clear patterns of constructions are disturbed in all languages by coordination of partial phrases. So what has to be acquired is rather some notion of dependency. Müller furthermore pointed out that there is interaction between various valence changing processes. The discussion in this paper will use the following examples:

- (3) a. dass er fischt
 that he.NOM fishes
- b. dass gefischt wird
 that fished is
- c. *dass er den Teich fischt
 that he.NOM the.ACC pond fishes
- d. *dass der Teich gefischt wird
 that the pond fished is
- e. dass er den Teich leer fischt
 that he.NOM the.ACC pond empty fishes
- f. dass der Teich leer gefischt wird
 that the.NOM pond empty fished is
- g. *dass er ihr fischt
 that he.NOM her.DAT fishes
- h. *dass er ihr den Teich fischt
 that he.NOM her.DAT the.ACC pond fishes

- i. dass er ihr den Teich leer fischt
that he.NOM her.DAT the.ACC pond empty fishes
‘that he fishes the pond empty for her’
- j. *dass der Teich gefischt wird
that the.NOM pond fished is
- k. dass ihr der Teich leer gefischt wird
that her.DAT the.NOM pond empty fished is
- l. dass sie den Teich leer gefischt bekommt
that she.NOM the.ACC pond empty fished becomes

The point about these examples is that there are phenomena like the benefactive and the resultative construction that extend the valence and that there are phenomena like the (dative) passive that reduce valence. Such phenomena can be covered in approaches with input and output relations as Müller (2018: Chapter 7) has shown. The accusative object *den Teich* ‘the pond’ of *leer fischen* ‘to fish empty’ is required, if the result predicate is present. Otherwise it is impossible, as (3c) shows. The accusative object can be promoted to subject in a passive construction only if the resultative predicate is present (3j) and (3f).

Haider (2024) discusses attested data that shows that a passive of sentences with dative passive is possible. If this is accepted, the following example has to be included as well:

- (4) dass der Teich leer gefischt bekommen wurde
that the.NOM pond empty fished become was
‘that somebody got the pond fished empty for him/her’

(4) is the *werden* passive of the dative passive in (3l).

Müller covered the data in (3) by a system of lexical rules and auxiliary verbs: the resultative lexical rule applies to an intransitive verb and licenses an additional result predicate and accusative object. The participle lexical rule blocks the designated argument (simplifying: the subject, Haider 1986) and depending on the auxiliary the accusative object or the dative object is promoted (Haider 1986, Müller 2002, 2003b). The benefactive lexical rule adds a dative to a transitive verb not taking a dative yet (Müller 2018: Chapter 7.2.3.1).

An early argument against phrasal phrase structure-based approaches like Generalized Phrase Structure Grammar (GPSG; Gazdar et al. 1985) was partial verb phrase fronting data like (5):

- (5) a. dass man den Wählern solche Märchen nicht
that one.NOM the.DAT voters such.ACC fairy.tales not
erzählen sollte
tell should
‘that one should not tell voters such fairy tales’

- b. Den Wählern Märchen erzählen sollte man besser
the.DAT voters fairy.tales.ACC tell should one.NOM rather
nicht.
not
'It is better not to tell voters fairy tales.'
- c. Den Wählern erzählen sollte man solche Märchen
the.DAT voters tell should one.NOM such.ACC fairy.tales
nicht.
not
'One should not tell voters fairy tales.'
- d. Märchen erzählen sollte man den Wählern nicht.
fairy.tales.ACC tell should one.NOM the.DAT voters not
'One should not tell voters fairy tales.'

GPSG used phrasal configurations for the analysis of argument structure constructions: lexemes with a certain number were inserted into phrase structure rules with the same SUBCAT number. For example, ditransitive verbs have the number 8 in Uszkoreit's (1987: 166) GPSG. So there is a phrase structure rule with a verb daughter with SUBCAT value 8 in which the verb is realized together with a nominative, dative, and accusative argument. Nerbonne (1986) and Johnson (1986) tried to reconcile this approach with partial fronting examples like those in (5) and succeeded, but they basically suggested a lexical encoding of valence: the solution they came up with was to redo valence in terms of features so that it can be ensured that exactly those arguments that are not realized in the fronted partial projection are realized in the so called Mittelfeld further to the right. They basically suggested a lexical encoding of valence parallel to what Categorical Grammar (Ajdukiewicz 1935, Steedman 2000) did and to what was later adopted in Head-Driven Phrase Structure Grammar (HPSG; Pollard & Sag 1994, Sag 1997, Müller et al. 2024).

The resultative constructions are also forming complex predicates and allow the same kind of partial frontings as complexes of non-finite verbs in German (Müller 2002: Chapter 5.1.14):

- (6) a. [Den Teich leer] sollte er diesmal nicht fischen.
the.ACC pond empty should he.NOM this.time not fish
- b. [Leer] sollte er den Teich nicht fischen.
empty should he.NOM the.ACC pond not fish
- c. [Leer fischen] sollte er den Teich diesmal nicht.
empty fish should he.NOM the.ACC pond this.time not
- d. [Den Teich leer fischen] sollte er diesmal nicht.
the.ACC pond empty fish should he.NOM this.time not

While such frontings are a problem for classical phrase structure-based approaches, TAG can cover these because of its extended domain of locality

(Joshi & Schabes 1997: 95–96). A TAG grammar will contain not just elementary trees with the verb, the result predicate, the object and the subject in their usual NP NP Pred V configuration, but also elementary trees in which the result predicate is fronted (6b), the result predicate and the verb is fronted (6c), while object and subject remain in the Mittelfeld and trees with the object and result predicate (6a) or object, result predicate and verb (6d) in fronted position. Resultative constructions take part in verbal complex formation. Vanilla TAG cannot describe certain constituent orders that arise with predicate complexes (Joshi, Becker & Rambow 2000). Therefore various extensions of TAG were suggested (Joshi et al. 2000, Lichte 2015). I do not want to enter this discussion here, which is the reason for not giving example trees for the analysis of (6). The solution will use a local tree containing the verb *fischen* ‘to fish’ and open slots for the result predicate, the accusative object and the subject. Hence, the partial projections will not be a fundamental problem for TAG.

The second argument against the GPSG approach was an argument from derivational morphology that interacts with valence (Müller 2016: Section 5.5.1). This argument also applies to current phrasal proposals (see for example Jackendoff’s (2011: 610) approach in *Simpler Syntax* and its discussion in Müller (2013b: 922–923)). Derivational morphology is discussed in Section 6.

The following section deals with inheritance as one of the important technical means used in constructionist theories. It will be shown that inheritance is not sufficient to capture phenomena interacting with valence.

3 Semantics, embedding, and inheritance

Müller (2006: Section 2.7) pointed out that it would be a considerable weakness of phrasal approaches if they had to write down all the possible configurations in (3) that are instantiations of the resultative construction. The general answer to redundancy problems and to questions regarding generalizations in Construction Grammar research is: inheritance hierarchies. But in terms of theory building and capturing generalizations this is not a convincing answer. Theories are supposed to derive facts not to describe a set of related phenomena as if they were unrelated. There have been suggestions in frameworks like TAG and HPSG that overcome such deficits by automatic type computation (Candito 1996, Koenig 1999). In such frameworks researchers can specify so-called partitions and one type belonging to every partition has to be chosen for a well-formed description of a certain object (Flickinger 1987: 20, Abeillé & Borsley 2024: 18). In Candito’s (1996) system there were three dimensions and from each one one maximal type had to be chosen. See also Kinyon et al. (2006). In more recent variants using XMG (Crabbé et al. 2013) all compatible types can be combined, but com-

bination can be blocked by stating incompatibility explicitly (Kallmeyer & Osswald 2013: 316, Lichte & Petitjean 2015: 199, Seyffarth 2023: 137).

There is a first problem with inheritance-based approaches: Semantics needs embedding (Müller 2006: 871, 2010: Section 4.2, 2017b). For example (7) is the meaning for (3e):

(7) `cause(fish(he), become(empty(pond)))`

While the meaning of the simple intransitive sentence (3a) is just `fish(he)`, the meaning of (3e) involves the cause operator, so the meaning of (3a) is embedded inside the meaning of the resultative construction.

Figure 4 visualizes the situation. There may be conflicting information

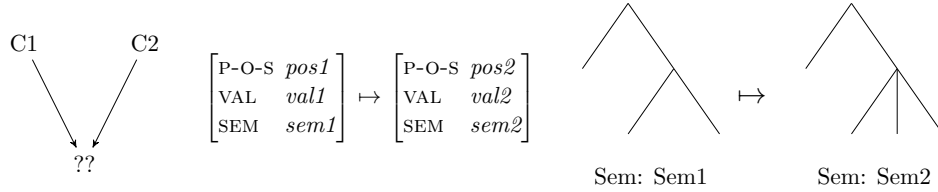


Figure 4: Inheritance vs. lexical rule vs. metarule

in an inheritance scenario (e.g., part of speech, valence or structure, and semantics). Semantic embedding is one possible reason for such a conflict. The alternatives to inheritance are lexical rules or metarules mapping trees to other trees. As shown in the figure, a mapping provides all the necessary flexibility. The part of speech information may be identical to the information of the input of the lexical rule or it can be a function of it. Similarly, the valence information may change and finally the semantics may be extended or embedded or left unchanged. The part of speech information together with valence information basically determines the tree structure a lexical head can appear in. TAG uses elementary trees directly, so the metarule depicted in the right figure maps a tree to another tree (of course with part of speech labels. These are not provided in the figure.). Again part of speech information and tree structure may change, as may the semantics. In particular the semantics of the input tree *Sem1* can be embedded under a newly added meaning component in the output tree.

One may try to do with inheritance and avoid lexical rules or metarules. There are tricks around the need of embedding. One can use so-called junk features in a feature value system to be able to refer to items that have to be embedded when new relations are added by inheritance (see Koenig 1999 and the critical discussion of this approach in Müller (2013a: Chapter 7.5.2.2, 2017b: Section 2.3). Goldberg’s (2003b) inheritance-based approach on Persian complex predicates is discussed by Müller (2010).). An alternative is to not use relations but use features. So instead of using *proposition(h1)* as common in some of the variants of Minimal Recursion

Semantics (Copestake et al. 2005, Ginzburg & Sag 2000: 368–369), one can just use a feature *MODE* with the value *proposition* (Sag & Wasow 1999: 107, Kathol 2000: 144, 170). This makes the information available without a relational representation. However, if several of such features are used, it cannot be recovered which relation scopes over which other one.

As was mentioned above, more recent versions of TAG use metagrammars to describe the elementary trees that constitute a TAG. Is the inheritance mechanism a game changer as far as valence is concerned? I suggest the answer is no. In what follows, I have a look at concrete proposals.

3.1 Caused Motion Construction as a clausal construction

Let us first have a look at the Caused Motion Construction, which is considered a subtype of the resultative construction by Goldberg & Jackendoff (2004: 540). Lichte & Kallmeyer (2017: 208) suggest the analysis in Figure 5.² A lexical verb can be inserted at $V_\diamond$. The subject is linked to

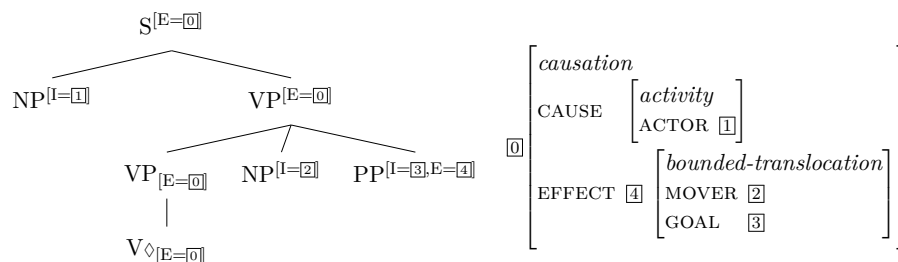


Figure 5: Caused-Motion Construction according to Lichte & Kallmeyer (2017: 208)

the actor, the object to the mover and the NP within the PP to the goal, the complete contribution of the PP is [4] and this is also the effect of the causation.

This approach and the earlier version by Kallmeyer & Osswald (2013) fails on sentences with weather verbs:³

(8) It rained the napkin off the table.

²A similar figure can be found in Kallmeyer & Osswald (2013: 310). The figure differs in that the E value of the lexical verb is not referred to and in that V and NP form a constituent and the object fills a theme role in the activity under CAUSE. I refer to the more recent figure in the following. The tree fragments provided in the inheritance hierarchy on p. 209 of Lichte & Kallmeyer (2017) combine a $V_\diamond$ and an NP into a VP. The PP is added in a tree in which a VP dominates a VP and a PP. The lower VP dominates a $V_\diamond$ (as in Figure 9 but with a PP instead of an AP). The tree that can be constructed from these components will be parallel to the tree shown in Figure 11.

³There do not seem to be any attested examples of expletives in Caused Motion Constructions or in the resultative construction in English. However, Felfe (2020: 256) discusses the German example in (i):

There is no actor and there is no object that is an argument of the main relation of the verb. The appropriate semantic representation for (8) seems to be:

(9) `cause(rain, off(napkin, table))`

The required semantic embedding can be reached in the TAG setting by taking the E of *rain* and identifying it with the value of CAUSE. But then we have an interesting new situation: The subject NP is not linked to anything in the tree. The revised analysis is shown in Figure 6. In lexicalized

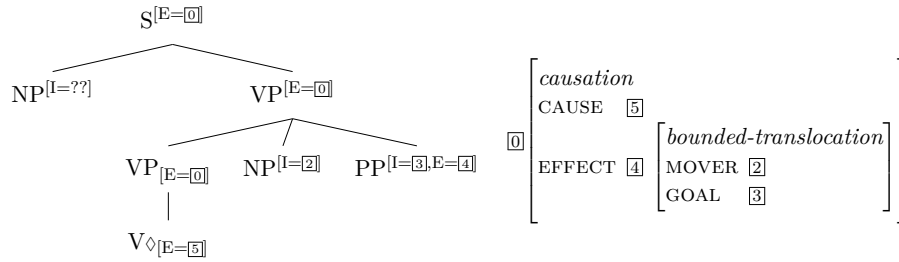


Figure 6: Caused-Motion Construction adapted for the treatment of expletive predicates

models like HPSG the information that the subject is an expletive lives in the lexicon. In TAG it lives in the trees (Figure 7). So either we have to

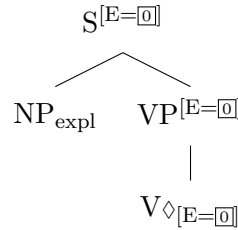


Figure 7: Tree for verbs with expletive subject

set up a system that gets the information about the expletive from the insertion place to the subject position in the tree or we have to assume that there are tree families for weather verbs. But then we have to combine these trees with the trees for Caused Motion Constructions. The normal trees for weather verbs project the complete semantics of the inserted verbs, while the Caused Motion trees have to insert the verb meaning under the cause

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- (i) [...] und am Ende hagelt es die Ernte kaputt.
 and at.the end hails it the crop to.pieces
 ‘and in the end it hails and destroys the crop’

The linking problems would have to be solved for German as well.

operator. There is no doubt that one can write down appropriate trees for all needed configurations but the properties of trees do not follow from unification of inherited parts. They have to be hand-crafted on a case by case basis. Of course one can split such handcrafted trees into components and combine them again, but this would overgenerate and it would be necessary to explicitly block unwanted configurations.

3.2 Caused Motion Construction as a VP construction

A more recent analysis was developed by Seyffarth (2023: 142). It is shown in Figure 8. The figure shows a verb phrase construction relating the con-

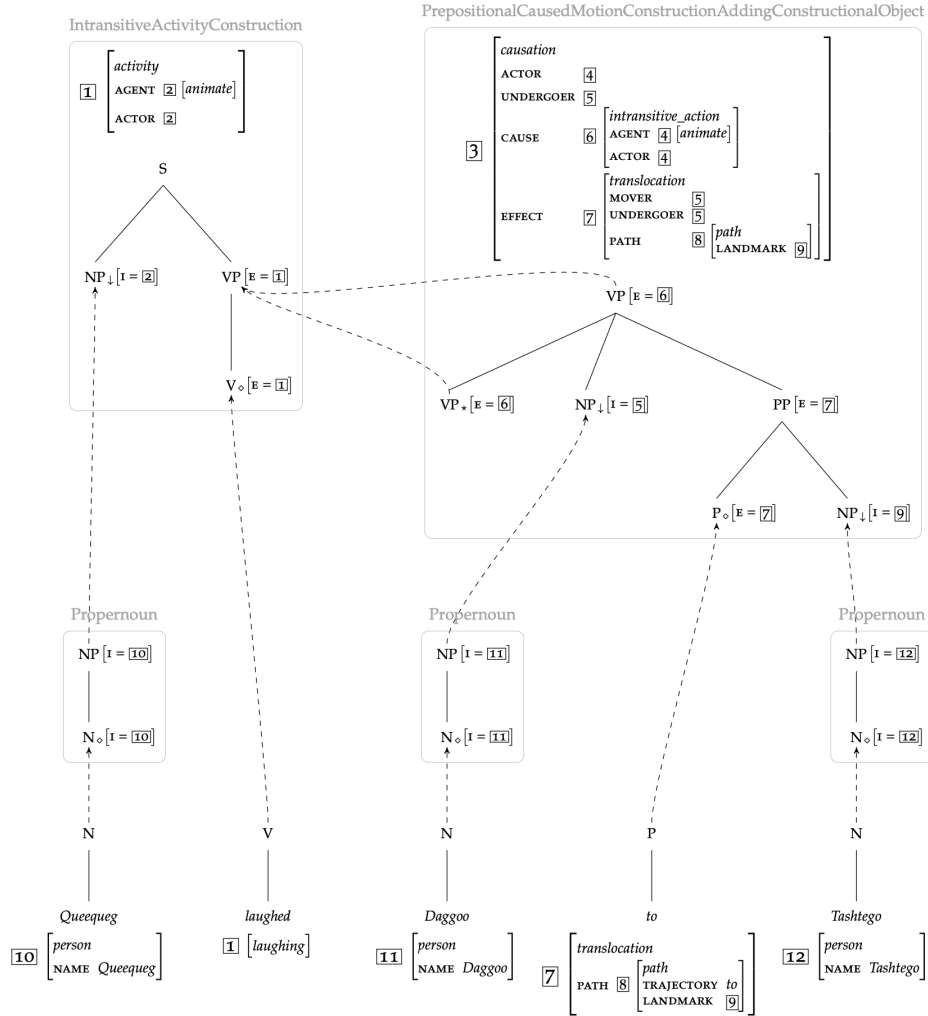


Figure 8: Prepositional Caused Motion Construction according to Seyffarth (2023: 142)

tribution of the verb [6] and the PP [7] as arguments of a *causation* relation. This Caused Motion Construction is adjoined to the VP node of an Intransitive Activity Construction. This has the potential to solve the problem with weather verbs since the Caused Motion construction can simply attach to any intransitive construction independent of the referential status of the subject. Of course the restrictions on [4] from within the Caused Motion Construction should be dropped. [3] should be the value of E at the upper VP node of the Caused Motion Construction. In this way it would be guaranteed that the semantics of the whole representation would be [3].

This works well but the question is how extendable this grammar/approach is. If we want to adopt this for the German data above, we would have to assume a phrasal construction introducing object, result predicate, and verb. This will give us (3e). But what about (3i)? One option would be to treat the combination of benefactive construction and resultative construction as a new construction that inherits from both the benefactive and from the resultative. But this is impossible, since the resultative semantics involves *cause'* and *become'*, while the semantics of the free dative contains the information that the referent of the dative NP is either the beneficiary or the one who suffers from the event described by the verb (Wegener 1985: 100). (10b) shows the meaning of (3i), repeated here as (10a):

- (10) a. dass er ihr den Teich leer fischt
 that he.NOM her.DAT the.ACC pond empty fishes
 ‘that he fishes the pond empty for her’
 b. cause(e1, fish(e2, he), become(e3, empty(s1, pond))) &
 benefactive/malefactive(e1, she)

I used event variables in (10) to be able to specify the event the *benefactive'* relation refers to. The highest operator in (10b) is the conjunction rather than the *cause'* relation. A way to model this would be to represent semantic contributions as sets and state constraints that relations have to be members of the respective sets.⁴ For this to work in inheritance networks, the number of elements in a set must be unspecified. This causes problems in similar approaches in HPSG (Müller 2013a: Chapter 7.5.2.2), but would work for TAG metagrammars, since grammar and metagrammar are different and one can decide to take the minimal model of the grammar that is described by the metagrammar. What can be modeled this way is relations that conjunctively refer to the same main variable. Embedding relations like *cause'* cannot be modeled by simply applying constraints on top of each other.

So, since different meanings are contributed by different additional constructions or construction parts (benefactive NPs, accusative NPs, and result

⁴Kallmeyer & Joshi (2003) use Minimal Recursion Semantics (Copestake et al. 2005), which is set-based.

predicates) we either have to change the representation of semantic information to sets or we have to use the only other means left: we have to assume that the Benefactive Construction is a phrasal construction in its own right. An adjunct construction, adding a dative. The benefactive dative can be combined with a two place verb with an accusative object:

- (11)
- a. dass er ihm den Koffer trägt
that he.NOM him.DAT the.ACC suitcase carries
'that he carries the suitcase for him'
 - b. dass er ihm die Miete bezahlt
that he.NOM him.DAT the.ACC rent pays
'that he pays him the rent'
 - c. dass er ihr einen Kuchen bäckt
that he.NOM her.DAT a.ACC cake bakes
'that he bakes her a cake'
 - d. dass er ihr das Laub verbrennt
that he.NOM her.DAT the.ACC leaves burns
'that he burns the leaves for her'
 - e. dass er ihm die Tür öffnet
that he.NOM him.DAT the.ACC door opens
'that he opens him the door'
 - f. dass er ihr die Serviette vom Tisch hustet
that he.NOM her.DAT the.ACC napkin off.the table coughs
'that she was affected by him coughing the napkin off the table'
 - g. dass es ihr die Serviette vom Tisch regnet
that it.EXPL.NOM her.DAT the.ACC napkin off.the table rains
'that she got the napkin rained off the table'

It follows that the fact that the verb the dative attaches to has a subject and an accusative object has to be visible for the adjoining dative construction. This means that either the adjoining verb has to be able to “see” all the parts and pieces of the transitive construction or that there is some valence representation at the verbal projection to which the benefactive construction adjoins. But this is exactly what I am arguing for in a lexical model: there has to be information about syntactic dependents available at crucial positions (the lexical head). Note that reference to the semantics of the verbal projection to which the dative attaches does not help, since it is not true that the dative attaches to a *cause* predicate or to a two-place predicate with actor and undergoer. As the examples in (11) show, there are various different options for semantic relations between the involved NPs. For example carrying a suitcase does not involve causation and raining does not involve an actor. So, the phrasal TAG model seems to be stuck in either an inheritance dilemma or an adjunction problem.

3.3 Constructing alternative trees from tree parts

A promising approach was suggested by Burkhardt et al. (2020). The authors suggest several tree fragments for the resultative construction in English that can be used to construct active and passive versions of the relevant types of resultative constructions.

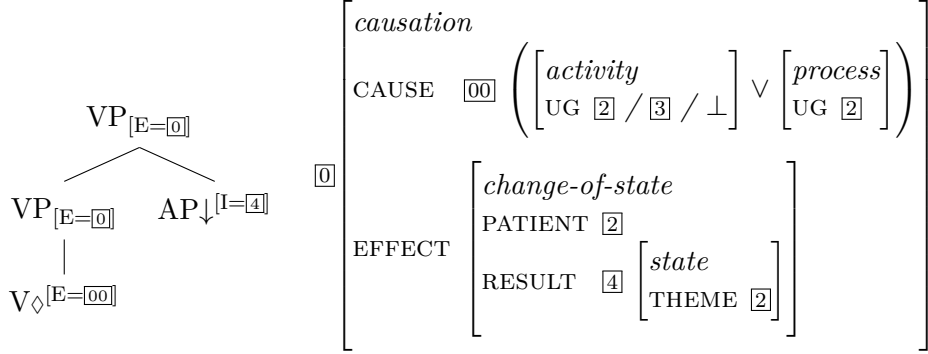


Figure 9: Resultative class in Metagrammar according to Burkhardt et al. (2020: 14)

The semantics is basically what I suggested above with the difference that the theme of the caused state is not involved in the activity or process that is causing the change of state and that *become'* does not have a PATIENT argument. The partial tree at the left side states that the resultative construction consists of a result predicate that is attached to the VP. Metagrammars allow to specify partial trees. So there can be an object in a full tree derived from the metagrammar, but trees without objects are possible too. The semantic embedding of the cause is taken care of by representing the semantic contribution of the lexical verb ($[00]$) as the value of CAUSE. Note though that this can only be done once. If we would want to add benefactive datives in the same way, this would not be compatible with the resultative construction. The benefactive meaning would have to scope over the *cause'* relation. We would have to stipulate a combined benefactive/resultative construction, which introduced both semantic contributions in one go or would have to introduce benefactives by adjunctions with the problems for dative passives discussed in the next section.

In addition to the class in Figure 9, Burkhardt et al. assume the three classes for subjects and objects in Figure 10. These fragments can be combined into trees for unergative/transitive verbs (**Actor-subject** $\wedge$ **Direct-object** $\wedge$ **Resultative**) and unaccusative verbs (**Undergoer-subject** $\wedge$ **Resultative**). Figure 11 shows the result of the combination with a direct object.⁵ Note that the conjunction is special in that the description lan-

⁵The $[5]$ probably refers to the tag of the actor role in **Actor-subject** in Figure 10. So, all occurrences of $[1]$ in Figure 10 should be replaced by $[5]$.

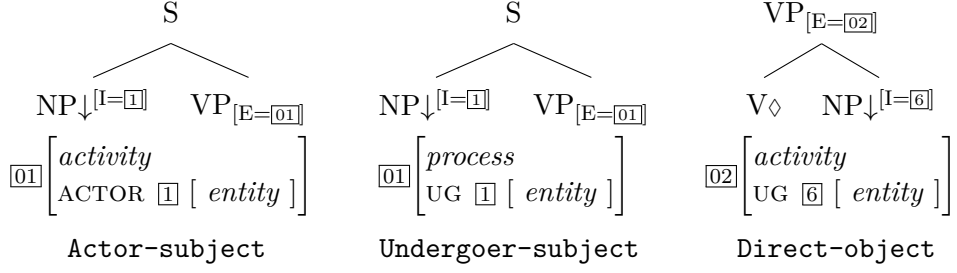


Figure 10: Subject and object classes according to Burkhardt et al. (2020: 15)

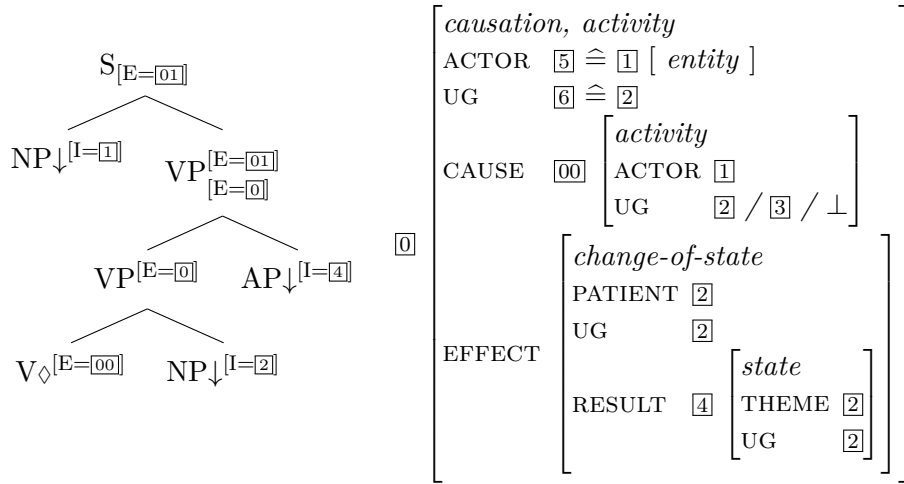


Figure 11: Resultative construction with subject and object according to Burkhardt et al. (2020: 16)

guage allows VPs (**Resultative** and **Direct-object** to be combined with S (**Actor-Subject**). This would not be possible within frameworks like HPSG, since S and VP are different symbols. The object VP is combined with one of the daughters of S.

The **ACTOR** and **UNDERGOER** features on the toplevel are contributed by the subject and object treelets. What is missing from simple unification of treelets is the connection between subject and object NP and the subject of the deeply embedded result state. The identification of all the **UNDERGOER** features and the introduction of these features into the respective AVMS is taken care of some general linking rules (Kallmeyer et al. 2016: 53-54). The rules that would be needed for the current example are: if there is a patient, it is also the undergoer; if there is a theme, it is also the undergoer; if there is an actor, it is the agent. The patient of a change-of-state is the patient of the result. If the cause of a causation has an actor, then this is also the actor of the entire causation. If the effect of a causation has an undergoer, then this is also the undergoer of the entire causation.

I argued for a representation using the *become'* operator above in (7). This operator is standard in such descriptions (Dowty 1979: 221, Van Valin 2005: 42). It does not have a patient/undergoer role. I think this is more adequate, since *become'* can embed predicates with expletive subjects (see also Müller 2002: 73):

- (12) In der Mensa wurde es laut.
 the the cafeteria became it loud
 'It got loud in the cafeteria.'

I think propagating semantic roles through semantic structures is a misuse of these structures to capture raising. The *cause'* relation should not be presented with an actor since expletive subjects are possible. The semantic representation that covers all cases is the one that I gave in (7) above. TAGs frame semantics structures are basically used like LFG f-structures (Bresnan 1982a). Bresnan provides an f-structure for *seem* which has a SUBJ feature that is identified with the SUBJ feature of the embedded predicate. See Figure 12. While LFG's f-structures look like semantic representations they are not. There is a dedicated semantic level that is derived from f-structures. Also: the subject is not a normal argument of SEEM-TO. The fact that it is raised is indicated by it being listed outside of the grammatical functions required by SEEM-TO.

This is what is done in frame-based TAG approaches: Semantic roles of embedded frames are made accessible to higher frames. Since they are also available at the top-most level, arguments at the sentence level can access and assign deeply embedded roles. With this move it becomes possible to state passive in the following way:

In passive voice, on the other hand, the undergoer becomes the

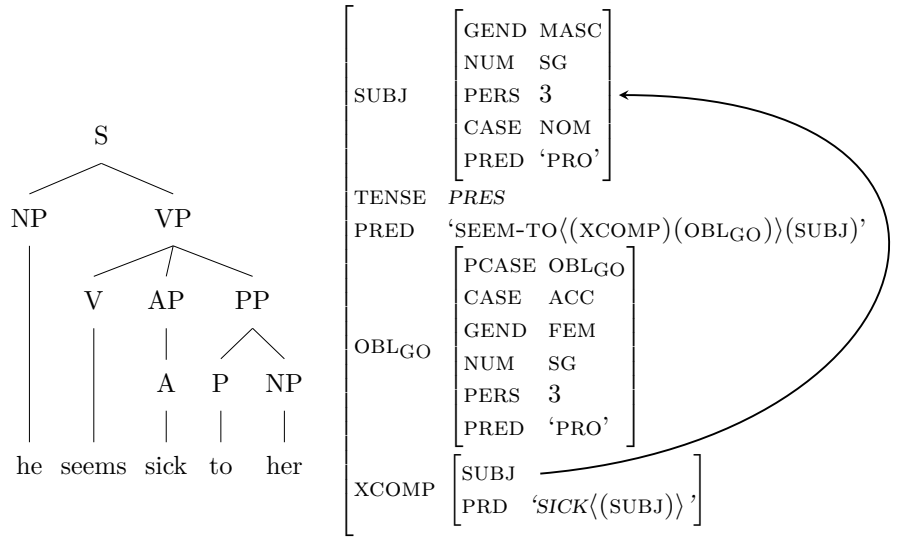


Figure 12: c-structure and f-structure for *He seems sick to her*. according to Bresnan (1982a: 362)

privileged syntactic argument while the actor may be optionally realized by a *by*-clause. (Kallmeyer et al. 2016: 52)

With a normal semantics for (13a) there would not be an undergoer for the *cause'* relation and the undergoer of the *fish'* relation is not realized.

- (13) a. He fishes the pond empty.
b. $\text{cause}(\text{e1}, \text{fish}(\text{e2}, \text{he}), \text{become}(\text{e3}, \text{empty}(\text{s1}, \text{pond})))$

Burkhardt et al. (2020: 19) refine the definitions for subjects in the way provided in Figure 13. With these definitions they can specify active and passive trees for the resultative construction with object undergoer:

- (14) Subject $\wedge$ Resultative $\wedge$
 $((\text{ActiveVerb} \wedge \text{Direct-object}) \vee \text{PassiveVerb}) \wedge$ (active)
 $(\text{PassiveVerb} \wedge \text{By-object}) \vee \text{PassiveVerb}$ (passive, by-PP)
 (passive, no by-PP)

While this analysis works for the cases discussed by the authors, it is just a complex statement of conjunctions and disjunctions that enumerates all possibilities: either we have an active verb with a direct object and resultative predicate or we have a passive verb without a direct object.

The analysis does not extend to the examples involving weather verbs with expletive subjects. The reason for this is that macro roles are assigned to referential arguments only (Van Valin & LaPolla 1997: 149). Verbs like *rain* do neither assign a semantic role nor a macro role (Van Valin & LaPolla 1997: 150).

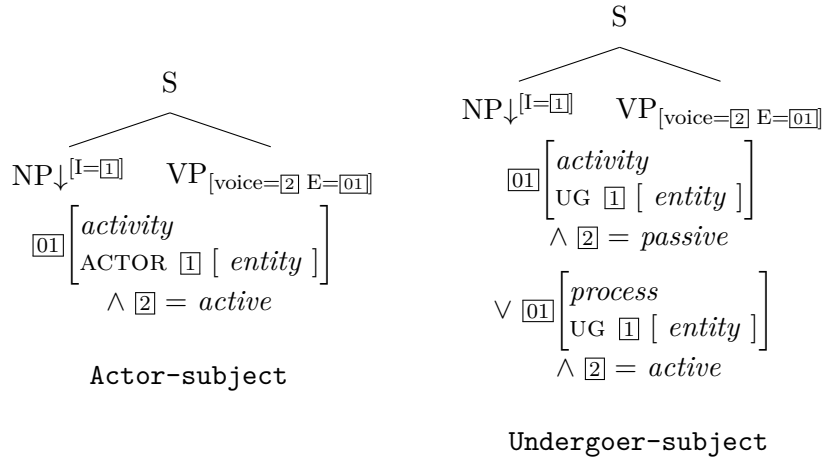


Figure 13: Subject linking for active and passive

4 Passives, datives and dative passives

As was shown in (3i) and (3l) the benefactive construction interacts with the resultative construction and the caused motion construction. The examples are repeated as (15) for convenience:

- (15) a. dass er ihr den Teich leer fischt
 that he.NOM her.DAT the.ACC pond empty fishes
 ‘that he fishes the pond empty for her’
 b. dass sie den Teich leer gefischt bekommt
 that she.NOM the.ACC pond empty fished becomes

The benefactives are so-called argument-adjuncts. They do not fill a semantic role of the verb, but supply additional information about an individual who benefits or suffers from a certain action.

The benefactive datives and other datives as well may be affected by the so-called dative passive. The dative element is then realized as subject of a clause with the auxiliary *erhalten/kriegen/bekommen* ‘get’ and the main verb in the participle form.

4.1 Benefactives as adjuncts

If the dative is not added in elementary trees but in adjunction structures, this results in problems with the passive. Let us consider the following data.

The dative in (16) is a body part dative (Engel 1977: 168–169). It is licenced by the presence of *Finger* ‘finger’.

- (16) a. daß wir noch nachsitzen mußten und auf die Finger
 that we.NOM still after.sit had.to and on the fingers

geklopft bekamen.⁶
 knocked got
 ‘that we still had detentions and that we got our knuckles
 rapped.’

- b. Jemand klopfte ihnen auf die Finger.
 someone.NOM knocked them.DAT on the fingers
 ‘Someone rapped them on their knuckles.’

The dative in (17) is also a body part dative related to the object in the Caused Motion Construction (Müller 2002: 132–133).

- (17) a. dass er die Seife aus den Augen gewaschen bekam
 that he.NOM the soap.ACC out the eyes washed got
 ‘that he got the soap washed out of his eyes’
 b. dass jemand ihm die Seife aus den Augen wäscht
 that someone.NOM him.DAT the soap.ACC out the eyes washes
 ‘that someone washes the soap out of his eyes’
 c. dass jemand die Seife aus den Augen wäscht
 that someone.NOM the soap.ACC out the eyes washes
 ‘that someone washes the soap out of the eyes’

The pure Caused Motion Construction is provided in (17c). (17b) shows the example with the added bodypart dative. (17a) is the dative passive of (17b). If these datives would be simply adjoined to trees, it would be unclear why (17a) is licensed. *die Seife* ‘the soap’ bears accusative. If *ihm* ‘him’ would be present in the active (17b) because of adjunction of the dative, it would be unclear why *die Seife aus den Augen gewaschen bekam* ‘the soap out of the eyes washed got’ should be licensed if *er* ‘he’ is not present. It is possible to require obligatory adjunction (Joshi & Schabes 1997: 74), but note that the requirement for dative passives is that there is some dative NP without any further restriction for the type of dative: it can be the object of a ditransitive verb like *schenken* ‘give as a present’, it does not have to be a free dative like a benefactive or a body part dative.

Note also that German datives do not get a macrorole Van Valin & LaPolla (1997: 354–355). So subject linking as suggested by Burkhardt et al. (2020: 19) would not work for dative passives, unless additional macroroles were stipulated.

4.2 Benefactives in elementary trees

The original promise was that TAG is well-suited to capture the ideas of phrasal Construction Grammar. What was suggested by Goldberg & Jackendoff (2004) was flat VPs with verb object and result predicate. What we have so far is a structured VP with the result phrase at a node dominating

⁶Frankfurter Rundschau, 03.06.1998, p. 2, quoted from Müller (2002: 150).

another VP node (Figure 11). It is unclear to me why this is needed. I think a totally flat analysis would be possible, since XMG allows to specify dominance constraints. So the **DirectObject** treelet could be combined with the **Resultative** treelet using the same dominating VP node. The result would be a leaner structure for resultative constructions. Figure 14 illustrates.

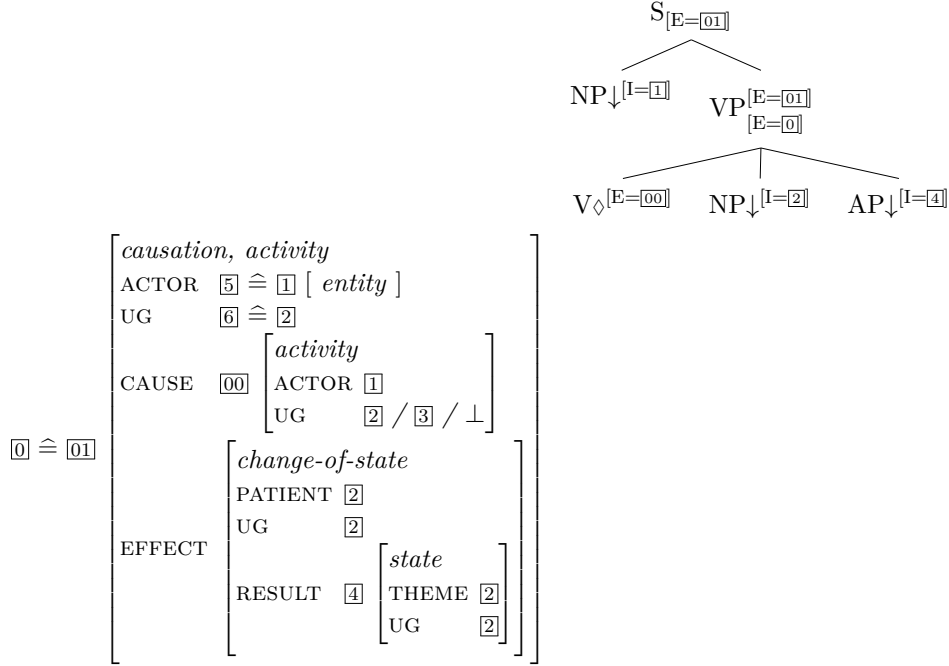


Figure 14: Resultative construction with subject and a flat VP

The important thing is that the VP node incorporates the semantics of the resultative construction. This is exactly what Goldberg envisioned: the verb is plugged in into the place for the lexical anchor and the resultative meaning is provided by the phrasal construction. There is a problem though: semantics needs embedding (Müller 2006: 871–872, 2010: Section 4.2). We need the projection from $V_{\diamond}$ to VP in order to introduce the resultative meaning. Once we used this projection, we cannot use it for anything else.

To see the problem more clearly, let us switch to German, where there is an interaction between the resultative construction and the benefactive. German is an SOV language without a dedicated subject position, so I assume the totally flat structure in Figure 15. Again the verb can be plugged into the $V_{\diamond}$ node, contributes its meaning via $[00]$ and the resultative meaning is represented at the S node as $[0]$.

The benefactive datives can appear with or without the resultative construction. (18) shows an example of the two-place verb *backen* ‘to bake’ that is used with a dative:

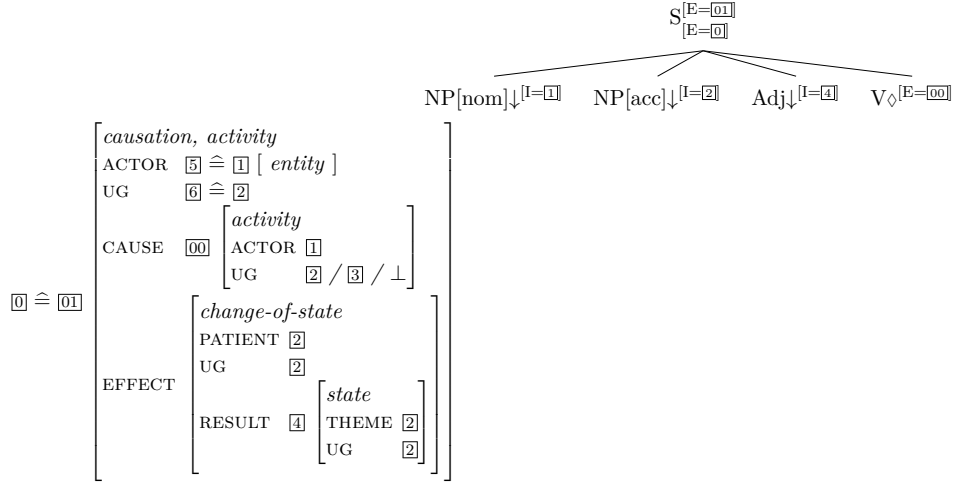


Figure 15: Resultative construction with a flat construction for German

- (18) [dass] er ihr einen Kuchen bäckt
 that he.NOM her.DAT a.ACC cake bakes
 ‘that he bakes her a cake’

The configuration is illustrated in Figure 16. If *bäckt* enters this construc-

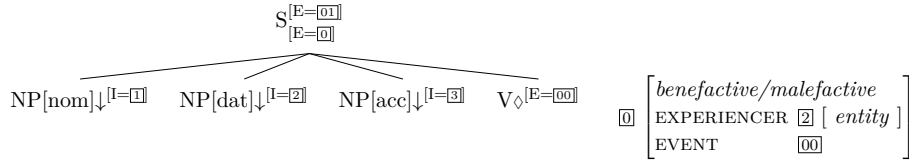


Figure 16: Benefactive construction with a flat construction for German

tion, its semantic contribution is identified with [00]. The *benefactive'* relation scopes over the *bake'* relation.

The solution to the problem of combining benefactive and resultative construction is the introduction of an extra node for the combination of adjective and verb. The resultative meaning is then added at this extra node and the benefactive meaning as contribution of the remaining construction. It is interesting to note that for German it would be sufficient to have a combination of adjective and verb. So instead of assuming the elementary tree in Figure 17, one could assume an elementary tree consisting of just the adjective and the verb. This would of course challenge the claim that one needs phrasal constructions involving subject, object, oblique, and verb to establish the form meaning pair, a point made in Müller (2007: 387), where I discussed the example in (19).

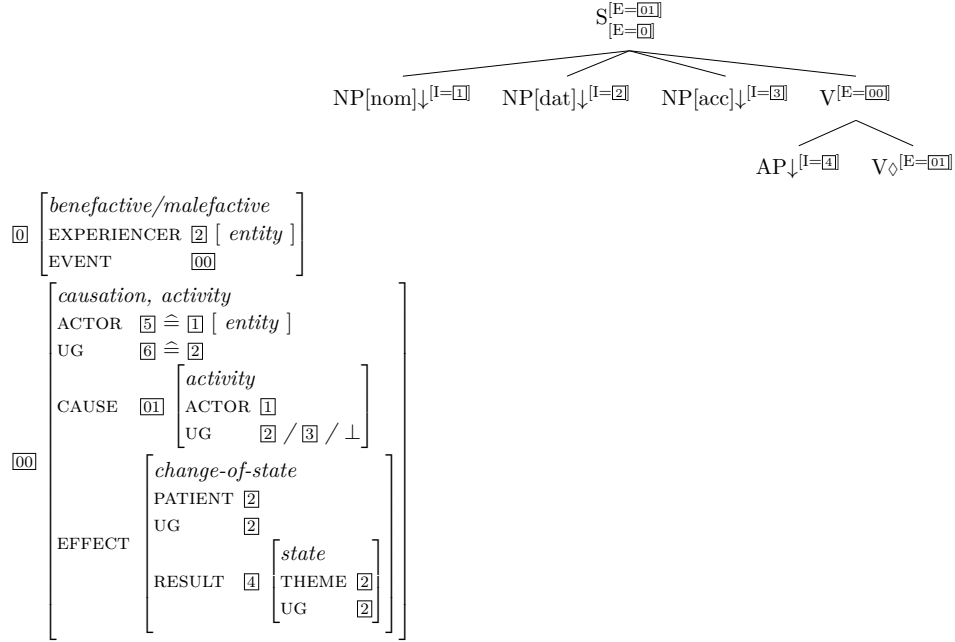


Figure 17: Benefactive construction with Resultative construction as predicate complex for German

- (19) Der kranke Mann wünschte sich, tot geschossen zu werden.
 the ill man wished SELF dead shot to be
 ‘The ill man wanted to be shot dead.’

(19) shows that there are situations in which the NP parts of the resultative construction are not visible, so the crucial parts of the construction seem to be the result predicate and the verb.

Burkhardt et al.’s (2020) approach can account for such examples since it uses an f-structure-like approach in the frame-semantics component: macroroles are propagated through the structure and hence the subject of *tot* ‘dead’, which is also the object of *tot geschossen* ‘shot’ and the subject of *tot geschossen zu werden* ‘to be shot’, is accessible for control. The macrorole approach fails on (20) under the assumption that datives do not get assigned macroroles:

- (20) Kim wünschte sich, den Teich leer gefischt zu bekommen.
 Kim wished SELF the.acc pond empty fished to get
 ‘Kim wished to get the pond fished empty.’

Note that this approach also works for cases with intransitive verbs or intransitively used transitive verbs. The reason for this is that what is inserted into the verb slot is the causative relation. The trick is the same: subject and object add *ACTOR* and *UNDERGOER* values to the *causation*

frame and the UNDERGOER may be shared with the UNDERGOER role of the verb ([2]) or the verb can have its own UNDERGOER ([3]) or no UNDERGOER at all ($\perp$).

One may wonder whether it could be possible to derive the English resultative constructions with standard TAG operations like adjunction and substitution, using an adjective-verb combination as in the analysis for German above. Unfortunately this is not possible since the object NP is serialized between verb and adjective. To license structures like the one in Figure 14, metagrammars are needed. A simple adjunction of a VP-AP combination to a VP node would not be sufficient, since the contribution of the V node is needed, since it has to be embedded under the *cause'* relation.

This leaves us with two radically different analyses of the resultative construction for English and German. Valence-based lexical approaches can capture both English and German with rather similar lexical rules. They just introduce a result predicate and a raised subject into a valence list (Simpson 1983, Verspoor 1997, Wechsler & Noh 2001: 399, Müller 2002: Chapter 5.2, 2018: Chapter 7.2.3.2, Butt et al. 2025: 23–24). Language-specific rules for SVO and SOV languages take care of the formation of VPs or complete clauses (Müller 2023a). Such lexical approaches have the additional advantage that they can deal with circumfixes in derivational morphology and hence do not have any problems with alledged bracketing paradoxes. This is dealt with in Section 6.

5 Iteration of valence changing processes

I wrote about the combination of the resultative construction and the benefactive construction, which are two argument adding constructions. They both interact with different forms of the passive (personal passive and dative passive), but there are also languages in which multiple argument reducing valence changes apply in succession. For example, the combination of the passive and the impersonal construction can be observed in languages like Lithuanian (Timberlake 1982: Section 5), Irish (Noonan 1994), and Turkish (Özkaragöz 1986). Özkaragöz’s Turkish examples are provided here with the original glossing as (21):

- (21) a. Bu şato-da boğ-ul-un-ur. (Turkish)
 this château-LOC strangle-PASS-PASS-AOR
 ‘One is strangled (by one) in this château.’
 b. Bu oda-da döv-ül-ün-ür.
 this room-LOC hit-PASS-PASS-AOR
 ‘One is beaten (by one) in this room.’
 c. Harp-te vur-ul-un-ur.
 war-LOC shoot-PASS-PASS-AOR
 ‘One is shot (by one) in war.’

Blevins (2003) provides an analysis of the interaction of passive and impersonal construction. Both the passive and the impersonal construction suppress the subject. In the passive the object is promoted to subject. In the impersonal nothing gets promoted. Note that the impersonal construction is not to be confused with the impersonal passive. The impersonal construction suppresses the subject and potential accusative objects can remain in the accusative (Blevins 2003: 495–496). The interesting fact is that the passive can interact with the impersonal construction and an argument that would be the object in an active clause and is the subject in a passive clause can be suppressed in an impersonal construction (Blevins 2003: 515). The interaction of the two phenomena cannot be covered in an inheritance hierarchy, since if we inherit from the passive construction, we commit to a certain structure. If we would inherit from the impersonal construction as well, we would get conflicting information since we need embedding: the impersonal construction has to apply to the element that has been passivized before. So the passivized structure has to be the input for the impersonal construction. Note that whatever tricks will be tried to fix this, a solution is not possible if it is based on inheritance, since if one inherits from two constructions A and B the result will always be $A + B$. $B + A$ will not be different from $A + B$ since inheritance from multiple supertypes results in structures for which all the constraints on the supertypes must hold. This is different with embedding constructions (lexical rules or metarules). If we have two rules A and B, they can be constructed such that B can apply to the output of A but not vice versa. This is exactly what is needed in the case of impersonals and passives: the impersonal rule can apply to the output of the passive, if the result of passivization provides a subject that can be suppressed, but passive, which is the suppression of the subject cannot apply to the impersonal construction, since the impersonal suppresses the subject without promoting anything to subject and hence the passive cannot apply, since there isn't any subject that could be suppressed.⁷ This

⁷One could fill the subject slot in an impersonal construction by an empty element. This would keep the passive structure in place and establish the impersonal construction by adding information. However, empty elements are frowned upon in Construction Grammar (Goldberg 2003a: Tenet 3) and the current goal of some members of the TAG community was to show that TAG is a good Construction Grammar. Note that there are two variants of empty elements in TAG. The first one assumes an elementary tree that is an empty element. This empty element can be inserted into other trees by substitution. The second kind of empty element lives in elementary trees only. Such empty elements are used for presentational issues only and do not play a role in derivations of complex linguistic objects (Lichte & Kallmeyer 2017: 208). The first type of empty elements is usually not assumed in TAGs. For the discussion at hand both types of empty elements are bad ones. It does not matter whether an empty element that is needed to make an analysis work is inserted during the construction of the tree or later by substitution. If one assumed empty elements for the suppression of the subject, one would have to find ways to rule out these elements in other NP slots. One would have to ensure that it does not enter subject slots in active sentences or passivized sentences but only those sentences that are instances of

is what phonologists and transformationalists discuss under the heading of feeding and bleeding (see p. 4 above) and what is sometimes claimed to be a disadvantage of lexical rules: one has to state a rule ordering (Crabbé et al. 2013: 596).

Of course it is possible to describe structures with successive applications of the passive and the impersonal construction, but this would be a description of the structures resulting from the successive application and it would not follow as the interaction of two independent constructions. To model this interaction is not possible in an inheritance hierarchy. It is straightforward in a system of lexical rules, though. As Blevins suggested: there is a lexical rule for the passive. It is paired with the passive morphology. The subject is suppressed and the remaining arguments are promoted, if there are any. The former object is the new subject. The lexical rule for the impersonal construction applies, adds the morphological marking, and suppresses the current subject, the former object. Of course one can develop a combined passive-impersonal construction with the morphological effects of both affixes and suppressing both arguments. But this would be an explicit restatement of the fact that the two constructions interact. The fact that the passivization of impersonal constructions is *not* possible would not follow from the theory. It would be possible to include the respective structure by just describing a structure that would be the result of applying the two constructions in the wrong order.

6 Morphology

I think the biggest problem for all phrasal approaches is the explanation of derivational morphology. First, German *-bar* ‘able’ derivation applies productively to verbs governing an accusative but not to verbs with dative objects (Riehemann 1998: 55).

- (22) a. Ich unterstütze den Mann.
I.NOM support the.ACC man
b. Der Mann ist unterstützbar.
the.NOM man is supportable
c. Ich helfe dem Mann.
I.NOM help the.DAT man
d. *Der Mann ist helfbar.
the.DAT man is helpable

The problem with derivational morphology is that it applies to stems. As is known since Krieger & Nerbonne (1993), derivational morphology cannot be done by inheritance, since embedding is needed for semantics and for changes

the impersonal construction.

of syntactic category. So the only other tool remaining is again combination using the standard TAG system of adjunction and substitution. Zinova (2021) developed a TAG analysis of Russian verbal prefixation, but what she did was combining verbs with additional trees for prefixes. Figure 18 shows an example. The prefix can adjoin to a VP node. A new, prefixed

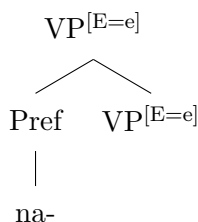


Figure 18: Russian prefix *ne-* according to Zinova (2021: 245)

VP node will result. This way one can change properties of the verb and one can add slots for further arguments. What one cannot do in TAG is getting rid of structure.⁸ Once the structure is there, we can insert material at certain nodes but we cannot change categories of nodes already present. So, if the claim is that the semantics of the resultative construction lives in a configuration consisting of at least verb, object, and result predicate, then there is no way of getting rid of one of these nodes, except the one of inserting an empty element, which hardcore versions of CxG avoid (Goldberg 2003a: Tenet 3).

- (23) a. dass der Anwohner den Teich leer fischt
 that the.NOM resident the.ACC pond empty fishes
 b. dass der Teich durch den Anwohner leerfischbar ist
 that the.NOM pond by the resident empty.fishable is
 c. die Leerfischbarkeit des Teiches durch den Anwohner
 the empty.fishability of.the pond.GEN by the resident

Figure 19 shows the tree configurations for the examples in (23). The left-most figure shows the configuration with subject, object, result predicate and verb. If this is the construction that links all the involved elements and provides the constructional meaning, it is unclear how the tree in the middle and the one at the right can be derived. The point is that the tree in the middle cannot be derived by automatic computation via inheritance since *leer fischbar* ‘empty fishable’ contains a modal semantics that embeds the semantics of *leer fisch-* ‘empty fish-’ (Gelhaus 1977: Section 3.1.2, Riehemann 1998: 68) and the category of a projection changes from verb to adjective. The usual way to treat this is that an adjectival affix attaches to

⁸One could fill slots by empty elements. See footnote 7 above on empty elements in TAG and Construction Grammar.

a verbal stem. But there is no TAG operation that can create the middle tree from the left-most tree by adjunction. The problem is that we have to replace the NP *der Anwohner* ‘the resident’ by a PP *durch den Anwohner* ‘by the resident’ and that *den Teich* ‘the pond’ has the wrong case. As usual, the arguments (NP and PP) are optional in nominal environments. Note that underspecification of case in the HPSG sense (Heinz & Matiassek 1994, Meurers 1999, Przepiórkowski 1999) will not help here. HPSG assigns case as soon as the environment in which an NP is realized is clear, but in the case of TAG the situation *is* clear: We are in a verbal environment in the left figure, in a different verbal environment in the middle figure and in a nominal environment in the right figure. Once we are at the level of TAG combinations we operate on full trees that could be complete utterances (unless they require obligatory adjunction). The case values of subject and object are fixed. This causes the problems with the morphological derivations. What one would need to relate the trees in Figure 19 is general, powerful transformations that map arbitrary trees into other trees as it was suggested by Crabbé et al. (2013: Section 3.1) for valence changing processes. Lexical rules as used in HPSG and SBCG or empty elements as used in Minimalism are an alternative (Müller 2023b: Chapter 19.5).

Note that the problem is even worse, since the structures in Figure 19 contain the adjective verb combination that is also part of the original construction, so there is some transparency in comparison with the construction originally stipulated. But morphologically a different structure is needed, namely the one that is provided in Figure 20. The nominalization using a circumfix shows that morphological material has to attach to the verbal stem. This results in a number of interesting bracketing paradoxes that have already been discussed with reference to German particle verbs (Bierwisch 1987: 165, Stiebels & Wunderlich 1994: 934, Stiebels 1996: 46, Lüdeling 2001: 106, Müller 2003a). The point is that *Ge-* *-e* attaches to the stem *fisch-* but scopes over the *cause'* relation. *Leergefische* refers to repeated annoying events of fishing something empty, that is, repeated annoying events of somebody causing something to become empty. It does not refer to repeated fishing events causing something to become empty. The parts of *Leergefische* that can be identified in the original phrasal resultative construction are marked in red in the figure. It is the adjective, the verbal stem and an NP in the wrong case. There is no way to derive the structure in Figure 20 from the left-most figure in Figure 19 by substitution or adjunction and there is no way to derive it by inheritance from a representation that contains a minimal configuration for resultative constructions including the subject or without subject. This is just technically impossible. Even the much reduced resultative construction just containing the adjective and the verb, which was suggested in Section 4.2 for German would not work here, since adjective and verb are not adjacent. This is different for valence-based approaches, since the selectional requirement for the adjective can be passed

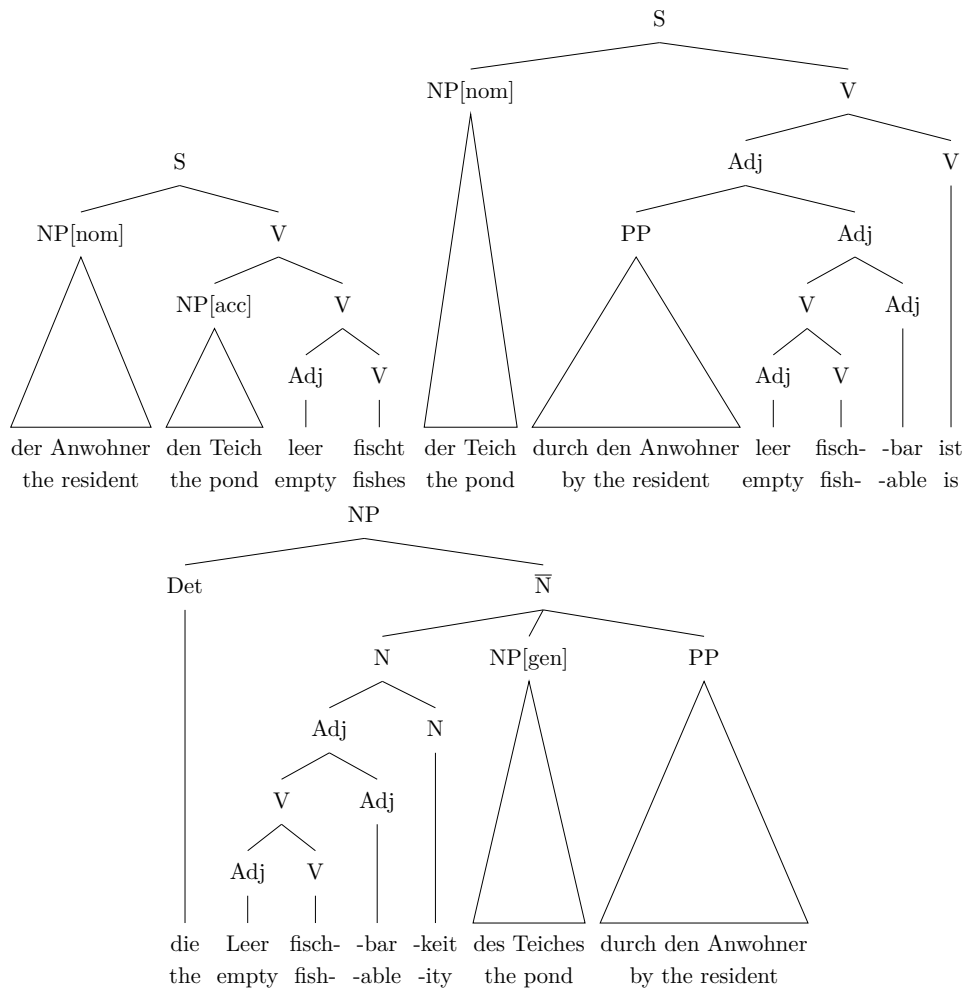


Figure 19: Trees for *dass der Anwohner den Teich leer fischt*, *dass der Teich durch den Anwohner leer fischbar ist*, and *die Leerfischbarkeit des Teiches durch den Anwohner*

down of linguistic observations and possible interactions between constructions or transformations of a power that was once common in Transformational Grammar but is long abandoned now.¹¹ Note that the rules that are used to relate trees in TAG are lexical rules or metarules rather than transformations. They do not relate complete trees with words or morphemes at the leaves. Rather they are relating elementary trees with open slots. In case of Lexicalized TAGs there is at least one leaf filled by a word and the rules relating trees would be lexical rules. In the case of non-lexicalized TAGs the rules would be metarules similar to the ones used in GPSG (Crabbé et al. 2013: 596). The difference is the greater domain of locality in TAG. GPSG just maps a phrase structure rule to another phrase structure rule. In TAG complex trees of the complexity discussed above can be mapped onto similar trees. As for GPSG grammars, the following holds for TAGs: during an actual analysis of a sentence, there is no need to apply transformations. The elementary trees are known and can be combined directly.

7 A simple solution using mappings to new linguistic objects

What is needed is a tool that maps one linguistic representation onto another. In early times of generative linguistics this was transformations. Later it was argued against transformations because of psycholinguistic implausibility (Bresnan 1978) and computational intractability (Kay 2011: 9–10, Peters & Ritchie 1973). A more suitable replacement was lexical rules¹² or lexical templates, as Goldberg (2013) prefers to call the lexical rules that are used today in HPSG (Briscoe & Copestake 1999, Meurers 2001, Davis & Koenig 2024: Section 5) and variants of Construction Grammar (Kay 2005, Sag, Boas & Kay 2012: Section 2.3). The alternative to such lexical rules are empty heads that are massively used in current Minimalist work (Müller 2023b: Chapter 19.5).

Lexical rules are relating lexical items with information about potential structure (Müller 2019) rather than relating objects with actual structure as in GB or transformational variants of TAG.

For the data described in this paper, the following lexical rules are needed:

1. a participle lexical rule blocking the designated argument (for perfect and passive; Haider 1986, Müller 2003b)

¹¹The power of TAG is restricted by the fact that the metarules can only apply to elementary trees, while in classical transformational grammar the transformations applied to arbitrary trees constructed by a phrase structure component. General transformational grammars have Turing power (Peters & Ritchie 1973), while TAGs correspond to mildly context-sensitive grammars (Joshi 1985).

¹²Dowty (1978) calls them *lexically governed transformations*.

2. a benefactive lexical rule adding the dative argument for verbs that take a nominative and accusative (Müller 2018: Chapter 7.2.3.1)
3. a resultative lexical rule adding an accusative object and a result predicate to a strictly intransitive verb (Verspoor 1997, Wechsler & Noh 2001: 399, Müller 2002: Chapter 5.2, 2018: Chapter 7.2.3.2, Butt et al. 2025: 23–24)
4. a *-bar* ‘able’ derivation lexical rule (Riehemann 1998, Müller 2003c: Section 5.2.2)
5. a *Ge-* *-e*-nominalization lexical rule (Müller 2002: Chapter 6.2.5.4, 2003c: Section 5.2.1)
6. a *-keit* nominalization lexical rule

Figure 21 shows the analysis in a lexicalist framework like HPSG that presents valence information in lists. A lexical rule applies to the verb stem *fisch-* and adds valence information and resultative meaning. The Benefactive Lexical Rule applies to the output and the resulting stem is inflected. There is only one possible order in which these rules can apply.

8 Generalizations and restrictiveness

The central claim of phrasal analyses within Construction Grammar is that certain meanings only come about since there is a certain configuration of constituents. Goldberg (1995: 152) argued that there is a phrasal configuration consisting of subject verb object and oblique and only if these parts of the construction appear together, we get the caused motion construction. This was repeated since then in the literature and two versions of this were discussed above: Lichte & Kallmeyer (2017) suggest a clausal structure with NP and VP and Seyffarth (2023) suggests a VP structure. The problem with these approaches is: If this combination of form and meaning is idiosyncratic and hence forms a construction, as is claimed by Goldberg (1995) and Goldberg & Jackendoff (2004),¹³ then this begs the question, how other related constructions are licenced? Why can the object be fronted? Why is there a passive? Culicover & Jackendoff (2005: Chapter 6.3) solved the problem of the passive by assuming different layers for grammatical functions and phrase structure and then assumed different mappings between these layers for active and passive. However, as Müller (2013b: 926–927) pointed out, this works only once. If you committed to one mapping, you are stuck. Phenomena like impersonals of passives (see Section 5) cannot be captured

¹³Goldberg (1995: 152) assumes the structure [SUBJ [V OBJ OBL]]. Goldberg & Jackendoff (2004: 534, 550) explicitly talk about VP configurations. On p. 540 they specify the syntax of constructions as NP V NP PP.

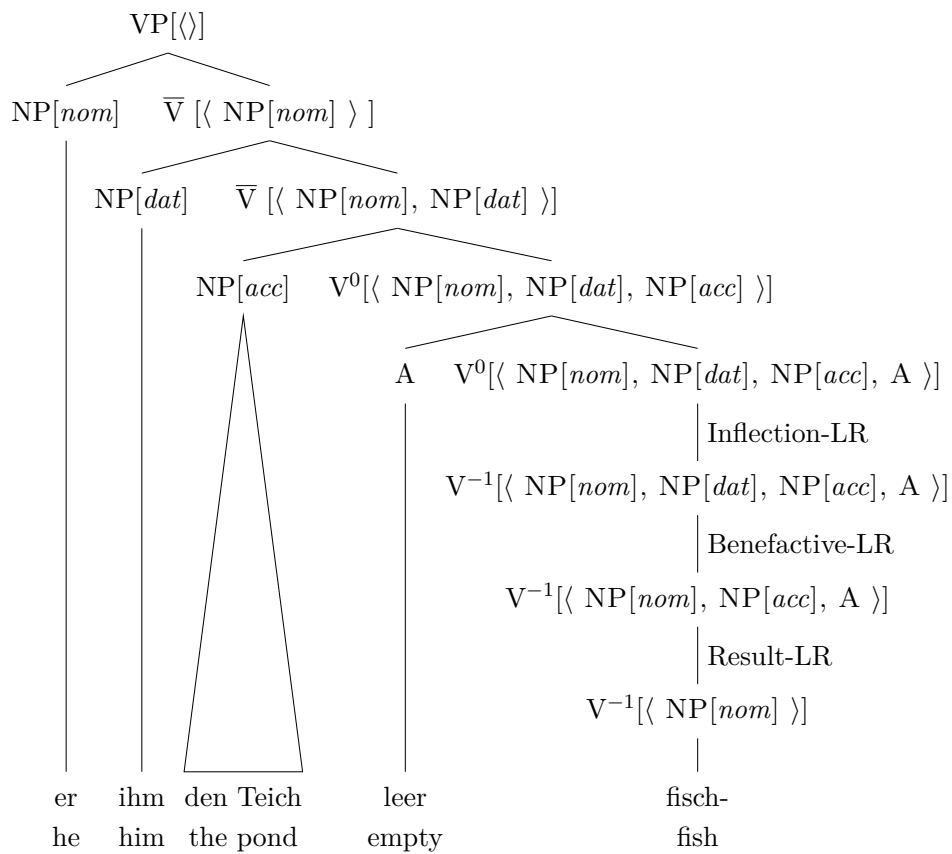


Figure 21: Analysis of *[dass] er ihm den Teich leer fischt* ‘that he fishes the pond empty for him’, an example in which the benefactive and the resultative construction interact

this way, since there is no possibility of modeling feeding relations between rules.

As for the Caused Motion Construction and Meta Grammars: there has to be a place in grammar in which it is stated that NP, V, NP, and PP together form the Caused Motion Construction with the respective meaning. So what has to be stated is that there is a certain dependency relation. The type hierarchy that is used to construct trees has to somehow use this knowledge.

In earlier versions of TAG, it was just stated that trees would adhere to the $\bar{X}$ theory (Kroch & Joshi 1985: 6). But in the end it was the (computational) linguist who just wrote down the trees and nothing prevented him or her from writing down any tree. There was no theory of the elementary trees. This changed with Vijay-Shanker & Schabes's (1992) work, who started to use inheritance networks for describing trees and lexical rules/metarules for relating them. Starting with Candito's (1996) paper, work on Metagrammars was intensified. Metagrammars are a means to describe trees and they are also inheritance-based. Since Metagrammars can be used in a way such that all types in the hierarchy are combined unless this is explicitly ruled out, there is some predictive power in these Metagrammars.

However, in other frameworks, the form of the trees is restricted by an explicit theory, namely phrase structure rules or dominance schemata (Jackendoff 1977, Bresnan 1982b: 10–11, Gazdar et al. 1985, Sag 1997). The dependency relation is stated in structures called *argument structure* in versions of HPSG and LFG. And then there are language (class) dependent principles mapping the elements in the ARG-ST list onto valence features. For example, the ARG-ST for a verb in the Caused Motion Construction would be:

(24) sneeze: $\langle \text{NP}[\textit{nom}], \text{NP}[\textit{acc}], \text{PP} \rangle$

The lexical item with the argument structure in (24) is derived from the lexical item for the intransitive verb *sneeze* by the following lexical rule:

$$(25) \begin{bmatrix} \text{HEAD} & \textit{verb} \\ \text{ARG-ST} & \langle \text{NP}[\textit{nom}] \rangle \\ \text{CONT} & \boxed{1} \end{bmatrix} \mapsto \begin{bmatrix} \text{HEAD} & \textit{verb} \\ \text{ARG-ST} & \langle \text{NP}[\textit{nom}], \text{NP}[\textit{acc}], \text{PP}:\boxed{2} \rangle \\ \text{CONT} & \begin{bmatrix} \textit{cause} \\ \text{CAUSE} & \boxed{1} \\ \text{EFFECT} & \boxed{2} \end{bmatrix} \end{bmatrix}$$

The semantic contribution of the intransitive verb ($\boxed{1}$) is embedded under the *cause* operator in the output of the lexical rule. The subject of the PP is the accusative object of the output verb (this is not shown in the rule). The semantic contribution of the PP ($\boxed{2}$) is identified with the effect of the *cause* operator. In our example, $\boxed{2}$ corresponds to off(napkin, table).

The arguments of the lexical item in (24) are mapped to the valence features by the very general Argument Realization Principle (Ginzburg & Sag 2000: 171) in the following way:

$$(26) \quad \begin{array}{cc} \text{English:} & \text{German:} \\ \left[\begin{array}{l} \text{SPR} \quad \langle \text{NP}[\textit{nom}] \rangle \\ \text{COMPS} \langle \text{NP}[\textit{acc}], \text{PP} \rangle \end{array} \right] & \left[\begin{array}{l} \text{SPR} \quad \langle \rangle \\ \text{COMPS} \langle \text{NP}[\textit{nom}], \text{NP}[\textit{acc}], \text{PP} \rangle \end{array} \right] \end{array}$$

Phrase structure rules relate elements in valence lists to positions in trees.

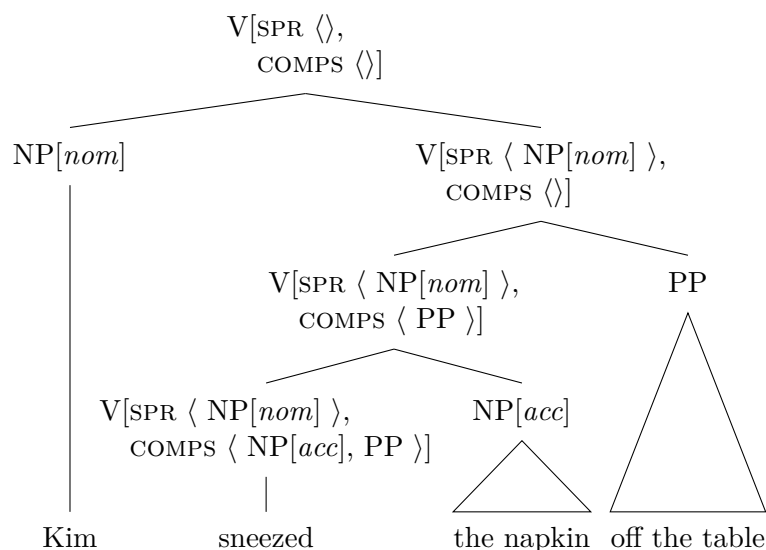


Figure 22: Caused Motion Construction in English

The schemata are really quite simple. They combine a head with one of its complements (27a) or with its specifier (27b):

$$(27) \quad \begin{array}{ll} \text{a. } H[\text{COMPS } \boxed{1} \oplus \boxed{3}] & \rightarrow H[\text{COMPS } \boxed{1} \oplus \langle \boxed{2} \rangle \oplus \boxed{3}] \quad \boxed{2} \\ \text{b. } H[\text{SPR } \boxed{1}] & \rightarrow H[\text{SPR } \boxed{1} \oplus \langle \boxed{2} \rangle] \quad \boxed{2} \end{array}$$

The symbol ‘ $\oplus$ ’ stands for append, which concatenates two lists. The COMPS list of the head in (27a) is split into three parts: two lists $\boxed{1}$ and $\boxed{3}$ and a third list with exactly one element $\langle \boxed{2} \rangle$. This element is combined with the head. The two lists $\boxed{1}$ and $\boxed{3}$ contain the complements that are not combined with the head. They are combined and the result $\boxed{1} \oplus \boxed{3}$ is the COMPS list of the mother node. The schemata in (27) do not say anything about the order of the daughters.

(27) is the generalization of all trees containing specifier, head, and complements. Adjuncts can be inserted at appropriate places of course. More specific constraints can be formulated for VO and OV languages. (28) shows the schemata for English and (29) for German:

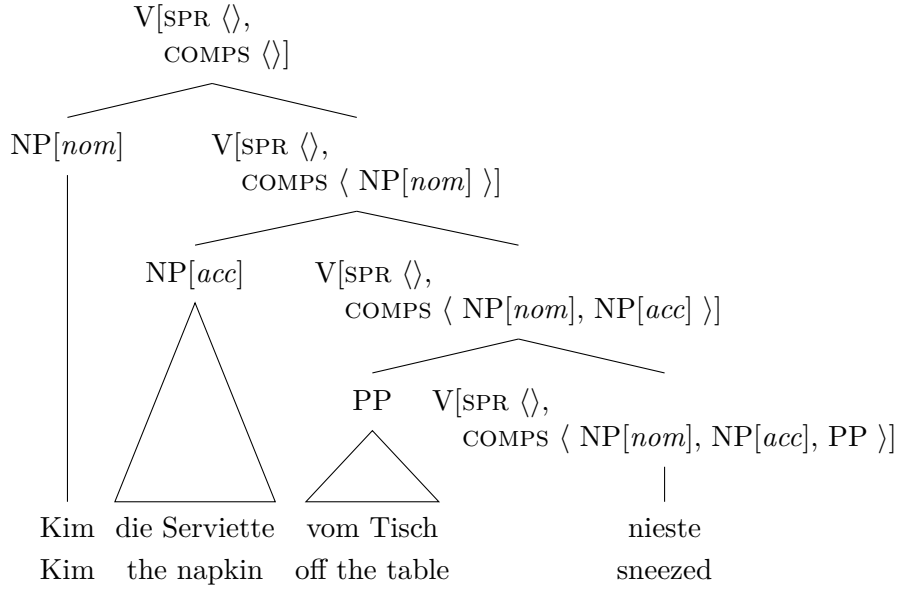


Figure 23: Caused Motion Construction in German

(28)

$$\begin{aligned}
 \text{a. } H[\text{COMPS } \boxed{3}] &\rightarrow H[\text{COMPS } \langle \boxed{2} \rangle \oplus \boxed{3}] \quad \boxed{2} & (\text{VO}) \\
 \text{b. } H[\text{SPR } \boxed{1}] &\rightarrow \boxed{2} \quad H[\text{SPR } \boxed{1} \oplus \langle \boxed{2} \rangle]
 \end{aligned}$$

(29)

$$\begin{aligned}
 \text{a. } H[\text{COMPS } \boxed{1} \oplus \boxed{3}] &\rightarrow \boxed{2} \quad H[\text{COMPS } \boxed{1} \oplus \langle \boxed{2} \rangle \oplus \boxed{3}] & (\text{OV}) \\
 \text{b. } H[\text{SPR } \boxed{1}] &\rightarrow \boxed{2} \quad H[\text{SPR } \boxed{1} \oplus \langle \boxed{2} \rangle]
 \end{aligned}$$

(28) and (29) contain ordering information already. The head precedes the complement in (28a) and it follows it in (29a). For (28a) $\boxed{1}$ is the empty list and hence omitted in the schema. Because of the threeway split in (29a), scrambling is accounted for. Depending on what is in $\boxed{1}$ and $\boxed{3}$ an element from the initial position, the middle or the end of the COMPS list can be combined with the head.

So the dependency information + very general syntactic rules that refer to the VO/OV classification of languages license concrete trees. For English, one gets the typical SVO order, for German one gets SOV with possible permutations of the arguments. At the level of argument structure it is captured that the Caused Motion Construction is parallel in German and English, despite the fact that the actual syntax of the two languages is quite different. The properties of lots and lots of trees are described by the schemata in (27) and the more specific versions in (28) and (29). The diversity comes about by inserting different lexical items into such trees. Of course this approach can also be assumed for Tree Adjoining Grammars. A prerequisite would be a lexical encoding of valence information. Lexical

rules could then be used to map intransitive verbs onto verbs entering the Cause Motion Construction and active verbs onto passive ones.

Concluding this section on metagrammars, trees, and generalizations, it can be said that metagrammars are used to describe the tree structures of TAG grammars. What is needed is a representation of the dependency between the involved parts of constructions and the meaning. This is done via argument structure representations in frameworks like LFG and HPSG. Some of these representations are valid for larger classes of languages although the actual structures in these languages varies (VO vs. OV). The form of trees can be described by very general dominance schemata.

9 Idioms and locality

Intuitively TAG seems to be the ideal framework for dealing with idioms, since one can specify trees that include all relevant parts of an idiom and the trees can be as complex as it is needed. Figure 3 already showed an example of the idiom analysis in TAG. But the same is true for Constructional HPSG as developed by Sag (1997) and used in the HPSG Handbook (Müller et al. 2024). Since HPSG models signs with phonological, syntactic, and semantic properties in one AVM and since mother-daughter configurations are included in these signs, HPSG can describe complex linguistic objects of the kind provided in Figure 3. Richter & Sailer (2009) showed how this can be done for idioms that were assumed to span clause boundaries.

Somewhat ironically, Sign-Based Construction Grammar (SBCG) was set up in a way to make assessing information about daughters in linguistic signs impossible. So describing idioms as phrasal configurations with partially fixed material, as it was done in TAG, is ruled out by design. Instead, idioms are covered via local selection (Sag 2007, Kay et al. 2025): a head selects a part of an idiom, which in turn may select other parts of the idiom. This way a dependency chain can be formed via local selections. Osborne & Groß (2012) independently claimed that idioms (phrasal constructions) can be and should be covered with dependency chains.

It seems to be an empirically open issue whether there are idioms that extend over clause boundaries. If this is indeed the case, trees of the complexity possible in TAG and Constructional HPSG would be needed. If it is not, the strict locality of SBCG would be sufficient. But of course the analyses of Constructional HPSG and TAG would still be valid.

In any case, one can conclude that there is no advantage of TAG over approaches like LFG or HPSG in terms of the treatment of idioms with partially fixed material.

10 Conclusion

I have shown that it is impossible to cover (certain interactions of) argument structure constructions in phrasal inheritance-based analyses within the framework of TAG unless one is willing to explicitly write down all possible configurations. If generalizations are to be captured, metarules for valence changes also in connection with derivational morphology seem to be inevitable. A metagrammar making use of inheritance does not help which can be shown with reference to mathematical properties of inheritance systems. So the conclusion seems to be that transformation-like metarules are needed and hence, that TAG is, despite all claims to the contrary, more similar to Transformational Grammars than to Construction Grammars.

A simple – constructional – alternative using lexical rules/lexical templates/lexical constructions has been suggested. Of course this analysis would also be available in a TAG setting. If the valence information of a lexical anchor determines into which tree a lexical element can be inserted (Kroch & Joshi 1985: 54–55), the approach with valence changing lexical rules can be used without any further modification.

As has been pointed out by Müller (2013b), the lexical rule-based system is also better in capturing crosslinguistic generalizations. This has been shown here again with respect to the benefactive construction and the resultative construction. The lexical rules for the benefactive and the resultative constructions work at least for German and English despite the fact that German is an SOV and English an SVO language. Facts about verb object order are independent of the respective constructions. For a purely surface-based view the fact that the constructions can be covered by the same rule in German and English must be surprising. For implemented grammars of English and German with resultative and benefactive lexical rules see the CoreGram project (Müller 2015).

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